

In Labs 2-4 of the ISSEM course you will apply the concepts you have learned in the theoretical modules on a real world application, namely the Infant Incubator, described separately.

Lab 4:

Coding Task

In this lab, fix/patch at least 4 of the identified vulnerabilities in Lab 3 (including the SQL Injection vulnerability).

1. Write a patch to fix the vulnerability in python/bash.
2. Write a test case to verify if the vulnerability is fixed. If possible, run your test cases and include "before and after" screenshots for each of your patches.

Also, provide brief explanations for any new functions you create.

Risk Assessment:

Update your Risk Assessment from Lab 3 with your findings from this Lab.

Submission:

1. Same format as Lab 3.

Notes:

How to write a test case

1. Test Case Description – What feature/behavior the test case is about e.g. "is authentication necessary for access"
2. Pre Condition – if there any conditions that must be true before the test runs (e.g. "the code does not include patch number xyz")
3. Expected Result – the conditions that are expected to be true after the test is run. (e.g. "access is granted")
4. Actual Result – the observable result of running the test, e.g. a trace showing that unauthenticated user is able to access resource A
5. Post condition – what should be true after the code is run e.g. "resource A is accessible to the unauthenticated user".
6. Pass/Fail – based on the expected result, did the test pass or fail (you should have two different results with a patch – otherwise the patch is ineffective).
7. Explanation: why does your test correctly check the feature/behavior that you are validating